MY THESIS TITLE

by

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Abstract

Your thesis abstract here. 350 words max.

Lay Summary

Your lay summary intended for the general public here. 150 words max.

Preface

This dissertation is original, unpublished, and independent work by Firstname Lastname with the following exceptions…

…

(If applicable!) No generative artificial intelligence tools were used in the research process, development, or writing of the dissertation.

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List of Symbols

This is an example list of mathematical symbols used. Examples assume $A \in \mathbb{R}^{I \times J \times K}$ is a third-order tensor, but notation and Julia code extends to any number of dimensions.

Name	Notation	Julia Syntax
First N Integers	$[N] = \{1, ..., N\}$	`1:N`
Array Element	$A[i, j, k]$	`A[i,j,k]`
(j, k) 1-Fibre	$A[:, j, k]$	`A[:,j,k]`
ith 1-Mode Slice	$A[i, :, :]$	`A[i,:,:]`
Transpose	$A^\top$	`A'`
Set of 1-Mode Slices	$\{A[i, :, :] \mid i \in [I]\}$	`eachslice(A; dims=1)`

List of Abbreviations

EMD	empirical mode decomposition
IMF	intrinsic mode function
UBC	The University of British Columbia

Acknowledgements

I am grateful for the funding and grant support from The University of British Columbia's Four Year Doctoral Fellowship and The Natural Sciences and Engineering Research Council of Canada's Canada Graduate Research Scholarship—Doctoral program…

I would like to thank my peer graduate students and postdoctorates…

Finally, I would like to thank my supervisory committee members and my supervisor…

To someone special.

1 Introduction

We can have *italicised* and **bolded** text by using _underscores_ and *asterisks*.

To type code, we can use `single backticks`. We can also specify the language with triple backticks like this ```typst #lorem(50)```. We'll use this lorem function for some filler text now.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

This is a non-numbered list using hyphens `-`.

- point 1
- next point
- something else

Here is a numbered list using plus signs `+`.

1. point 1
2. next point
3. something else

Here's a block quote.

> "I am inside a block quote. You usually only use these for longer quotes that span longer than a single line, or to emphasize a particular point."

This text continues without the indentation from the quote.

This is a sentence with a footnote.[1]

Here is an inline equation $a^2 + b^2 = c^2$. And here is a display equation with some text using the function `mathtext("the text")`

$$\min_x f(x) \quad \text{s.t.} \quad x \geq 0. \tag{1.1}$$

This ensures text is properly italicised when used in theorem environment like in Definition 1.1.

[1] This text is in the footnote. They can include equations like this, $\varepsilon > 0$, among other things.

Definition 1.1 (An Optimization Problem): *A general nonnegative optimization problem is written*

$$\min_x f(x) \quad \textit{s.t.} \quad x \geq 0.$$

Here's a link to a website. This one links to the GitHub repository for BlockTensorFactorization.jl.

This is an abbreviation for The University of British Columbia (UBC). Every time it is used after the first, it will be short by default: UBC. You can click it to see the list of abbreviations!

Now we reference multiple sections at once with `#Cref[@ref1 @ref2 @ref]`: see Sections 1, 2 and 4. This function also works with tables and figures: Figures 1 and 2. To reference multiple equations, use `#eq_ref[@eq1 @eq2]`: (Equations 1.1 and 3.1). Use `#thm_ref[@thm1 @thm2 @thm3]` to reference multiple theorems, lemmas, definitions, etc. as long as they are the same type: Definitions 1.1, 2.2, and 2.3. We need custom functions to properly include the section number in equation and theorem references.

To end a line early, use a backslash `\` like this.
Now this is on the next line. Note there is extra space at the end of the line, but we did not start a new paragraph.

1.1 Contributions

In this thesis, we provide new ways to ...

Now we will force a page break with `#pagebreak()`.

This text starts on the next page. Notice the running headers above only appear on pages after a new level 1 section.

1.2 Roadmap

The high-level flow of this thesis starts with…

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut.

2 Literature Review

Beginning with our notation, we review the necessary background ...

2.1 Some Definitions and Theorems

We state the usual definition for a convex function in Definition 2.1, and a standard result in Proposition 2.1 immediately afterward.

Definition 2.1 (Convex Function): *A function $f : \mathcal{D} \subseteq \mathbb{R}^I \to \mathbb{R}$ is convex when*

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y),$$

for all $0 \leq \lambda \leq 1$ and $x, y \in \mathcal{D}$.

Proposition 2.1 (Differentiable Convex Functions [1 (Thm. 25.1)]): *Let f be differentiable.*

$$f \text{ is convex} \quad \iff \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \forall x, y \in \mathcal{D}.$$

We cite Rockafellar right in the name of Proposition 2.1 and reference the specific theorem it comes from.

Here are two more definitions shown in Definitions 2.2 and 2.3.

Definition 2.2 (Lipschitz Function): *A function $f : \mathcal{D} \subseteq \mathbb{R}^N \to \mathbb{R}$ is L_f-Lipschitz when*

$$|f(a) - f(b)| \leq L_f \|a - b\|_2, \quad \forall a, b \in \mathcal{D}.$$

We call the smallest such L_f the *Lipschitz constant of f.*

Definition 2.3 (Functional Infinity Norm): *The infinity norm $\|f\|_\infty$ of a function $f : \mathcal{D} \subseteq \mathbb{R}^N \to \mathbb{R}$ is the largest magnitude producible by f's output,*

$$\|f\|_\infty = \sup_{A \in \mathcal{D}} |f(A)|.$$

We call a function bounded *if its infinity norm is a finite number.*

Here is a lemma that is immediately followed by its proof.

Lemma 2.1 (Product of Lipschitz Functions): *Let $f, g : \mathcal{D} \subseteq \mathbb{R}^N \to \mathbb{R}$ be bounded and L_f- and L_g-Lipschitz functions respectively.*

Then, the product function $(fg)(A) = f(A)g(A)$ is L_{fg}-Lipschitz with constant

$$L_{fg} = L_f \|g\|_\infty + L_g \|f\|_\infty.$$

Proof of Lemma 2.1: Let $f, g : \mathcal{D} \subseteq \mathbb{R}^N \to \mathbb{R}^M$, and $a, b \in \mathcal{D}$. Then

$$\begin{aligned}
|f(A)g(A) - f(B)g(B)| &= |f(A)g(A) - f(A)g(B) + f(A)g(B) - f(B)g(B)| \\
&\leq |f(A)||g(A) - g(B)| + |g(B)||f(A) - f(B)| \\
&\leq |f(A)|L_g\|A - B\|_F + |g(B)|L_f\|A - B\|_F \\
&\leq \|f\|_\infty L_g\|A - B\|_F + \|g\|_\infty L_f\|A - B\|_F \\
&= \left(\|f\|_\infty L_g + \|g\|_\infty L_f\right)\|A - B\|_F. \qquad \square
\end{aligned}$$

Theorem 2.1 has its proof in the appendix under Appendix A.2. This theorem uses `#box` to force "$a, b \in \mathbb{R}^N$" to be on the same line. We also add a label to Equation 2.1 so we can reference it.

Theorem 2.1 (Lipschitz Function Interpolation): *Let $f : \mathbb{R}^N \to \mathbb{R}$ be L_f-Lipschitz. For all $a, b \in \mathbb{R}^N$ and $\lambda_1, \lambda_2 \geq 0$, $\lambda_1 + \lambda_2 = 1$, the error between f evaluated at a convex combination of a and b and the linear interpolation of $f(a)$ and $f(b)$ satisfies*

$$|f(\lambda_1 a + \lambda_2 b) - (\lambda_1 f(a) + \lambda_2 f(b))| \leq 2L_f\lambda_1\lambda_2\|a - b\|_2. \tag{2.1}$$

Proof of Theorem 2.1: See Appendix A.2. $\square$

We also have corollaries, assumptions, and remarks. We can even reference Corollary 2.1 and Assumption 2.1.

Corollary 2.1 (Something Obvious): *This follows from Theorem 2.1*

Assumption 2.1: *We live in a world.*

Remark: Just pointing this out.

2.2 Citations

We can string a bunch of citations that will be nicely collapsed automatically by our citation style. [2–5]. Some of them can include references to specific sections [4 (Sec. 1), 5].

2.3 More Literature Review with a Figure

One family of methods separating signals into intrinsic mode functions (IMFs) which are amplitude-frequency modulated sinusoids $s(t) = a(t)\sin(\phi(t))$ (or their complex version, $a(t)e^{i\phi(t)}$)

with instantaneous amplitude $a(t) > 0$ and frequency $\phi'(t) > 0$. Figure 1 shows a sketch of three IMFs. One tool in this category is empirical mode decomposition (EMD) [6].

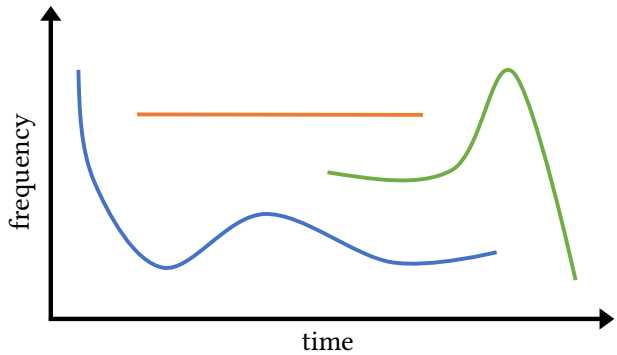

Figure 1: Example sketch of a signal with three IMF sources. This is a longer caption that only appears in the main body of the text.

EMD suffers from "mode mixing" and sensitivity to noise among other issues, so improvements have been proposed to combat these issues [7, 8].

3 Display items

3.1 Tables

Here, Table 1 shows the convergence criteria implemented in v0.4.0 of the tensor factorization package BlockTensorFactorization.jl [9].

Table 1: Convergence criteria implemented in BlockTensorFactorization.jl.

Criteria	Definition
`ObjectiveValue`	$f(X^t)$
`ObjectiveRatio`	$f(X^{t-1})/f(X^t)$
`IterateNormDiff`	$\left\|X^t - X^{t-1}\right\|_F$
`IterateRelativeDiff`	$\left\|X^t - X^{t-1}\right\|_F / \left\|X^{t-1}\right\|_F$
`RelativeError`	$\|X^t - Y\|_F / \|Y\|_F$
`GradientNorm`	$\|\nabla f(X^t)\|_F$
`GradientNNCone`	$d(-\nabla f(X^t), N_{\geq 0}(X^t))$

3.2 A table with custom formatting

Table 2 shows the behaviour of randomizing updates in BlockTensorFactorization.jl. We use the following typst code to colour `true` green and `false` red to make it easier to read.

```
#show raw: it => {
  if it.text == "true" {
    set text(fill: color.rgb("#3a8027"))
    it
  }
  else if it.text == "false" {
    set text(fill: color.rgb("#a42929"))
    it
  }
  else {
    it
  }
}
```

Table 2: Full description of randomizing the order of updates within a `BlockedUpdate`. The headers Group, Random, and Recursive stand for the keyword arguments `group_by_factor`, `random_order`, and `recursive_random_order`.

Group	Random	Recursive	Description
`false`	`false`	`false`	In the order given
`false`	`false`	`true`	Randomize how existing blocks are ordered (recursively)
`false`	`true`	`false`	Randomize updates, but keep existing blocks in order
`false`	`true`	`true`	Fully random
`true`	`false`	`false`	In the order given
`true`	`false`	`true`	Updates for each factor are in a random order
`true`	`true`	`false`	Random order of factors, preserve updates in each factor
`true`	`true`	`true`	Random, except updates for each factor are done together

3.3 Code blocks

To constrain a vector $v \in \mathbb{R}^J$ to the simplex

$$\Delta_J = \left\{ v \in \mathbb{R}_+^J \;\middle|\; \sum_{j=1}^{J} v[j] = 1 \right\},$$

we can apply a Euclidean projection

$$v \leftarrow \operatorname*{arg\,min}_{u \in \Delta_J} \|u - v\|_2^2.$$

The Euclidean simplex projection can be done with the following implementation of Chen and Ye's algorithm [10] in BlockTensorFactorization.jl [9]. The essence of the algorithm is to efficiently compute the special $t \in \mathbb{R}$ so that

$$v \leftarrow \max(0, v - t\mathbb{1}) \in \Delta_J. \tag{3.1}$$

```
function projsplx!(v)
    n = length(v)
    y_sorted = sort(v[:]) # Vectorize and sort all entries
    total = y_sorted[n] # Initialize total
    i = n - 1
    t = zero(eltype(v)) # Ensure t has scope outside the while loop

    while true
        t = (total - 1) / (n-i)
        y_i = y_sorted[i]
        if t >= y_i
            break
        else
            total += y_i
            i -= 1
        end
        if i >= 1
            continue
        else # i == 0
            t = (total - 1) / n
            break
        end
    end

    @. v = ReLU(v - t)
    return v
end
```

3.4 Grid of figures

Figure 2 shows an example grid of figures.

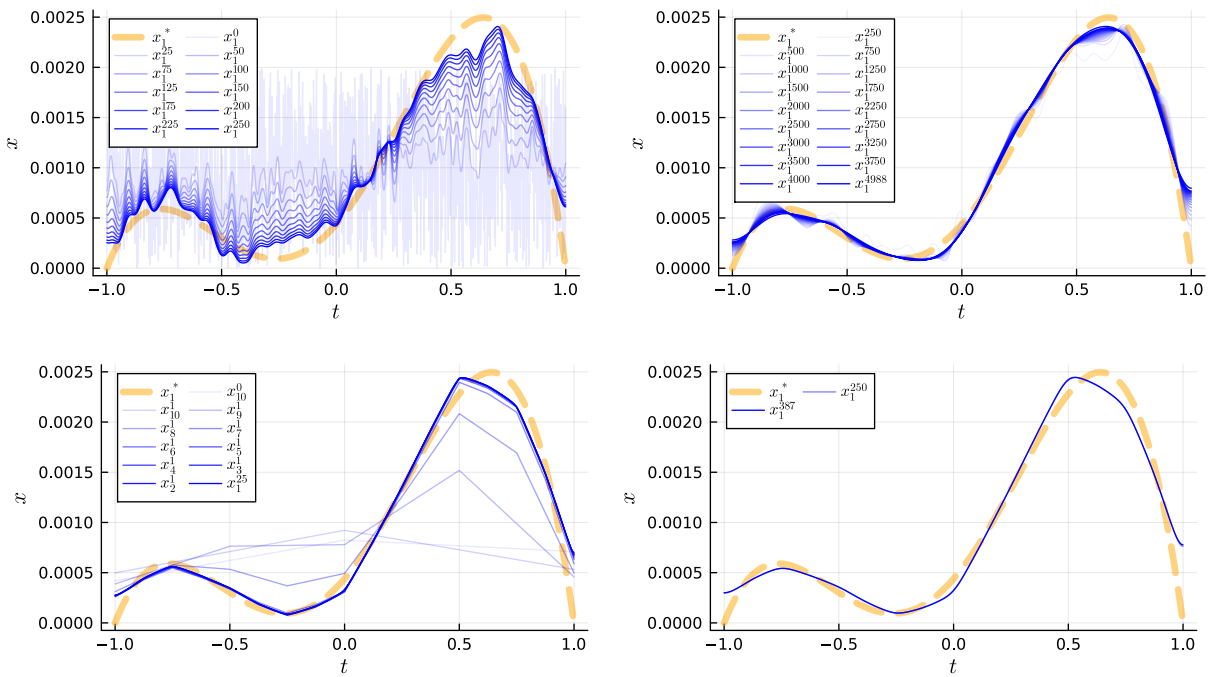

Figure 2: Here is an example grid of figures. These are taken from Richardson, Marusenko, and Friedlander [11, 12].

3.5 Algorithms

Algorithm 1 is taken from [11]. It references P_s which is the following equation with a custom label,

$$\min_{x_s}\{\tilde{\mathcal{L}}_s(x_s) \mid x_s \in \tilde{\mathcal{C}}_s\}. \tag{P_s}$$

Algorithm 1: The Multiscale Method. Line 6 specifies Greedy or Lazy versions.

1: Randomly initialize x_s^0 at the coarsest scale $s = S$.
2: Perform K_s iterations of projected gradient descent on x_s^0 to approximately solve P_s at scale $s = S$ and obtain $x_s^{K_s}$.
3: **for** scales $s = S-1, S-2, \ldots, 1$ **do**
4: Interpolate $x_{s+1}^{K_{s+1}}$ to obtain $\underline{x}_{s+1}^{K_{s+1}}$.
5: Initialize the next finest scale $x_s^0 = \underline{x}_{s+1}^{K_{s+1}}$ using the previous scale's interpolated solution.
6: Perform K_s iterations of projected gradient descent on x_s^0 (Greedy) or $x_{(s)}^0$ (Lazy) to approximately solve P_s at scale s and obtain $x_s^{K_s}$.
7: **end**
8: **return** Approximate solution $x_1^{K_1}$ for P_s at $s = 1$

4 Conclusion

This thesis explores... `some text`

4.1 Discussion

Some things to say...

4.2 Next Steps

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[9] N. Richardson, N. Marusenko, and M. P. Friedlander, "BlockTensorFactorization.jl v0.4.0," 2025. https://github.com/MPF-Optimization-Laboratory/BlockTensorFactorization.jl.

[10] Y. Chen and X. Ye, "Projection Onto A Simplex," Feb. 2011. http://arxiv.org/abs/1101.6081.

[11] N. J. E. Richardson, N. Marusenko, and M. P. Friedlander, "Multiple Scale Methods For Optimization Of Discretized Continuous Functions," Dec. 2025. http://arxiv.org/abs/2512.13993 (accessed 16 Jan. 2026).

[12] N. Richardson, N. Marusenko, and M. P. Friedlander, "richardson-multiscale-2025," 2025. https://github.com/MPF-Optimization-Laboratory/richardson-multiscale-2025.

Appendices

Appendix A: This is the First Appendix

A.1 This text is rich and can include references to Section 1 and code `AbstractMatrix`

We may want to use the above function to reference proofs of particular theorems like in Appendix A.2.

A.2 Proof of Theorem 2.1 (Lipschitz Interpolation)

Let $f : \mathbb{R}^N \to \mathbb{R}$ be L_f-Lipschitz, $a, b \in \mathbb{R}^N$ be vectors, and $\lambda = \lambda_1$ so that $1 - \lambda = \lambda_2$ for $\lambda \in [0, 1]$. Our goal is to bound

$$|((1-\lambda)f(a) + \lambda f(b)) - f((1-\lambda)a + \lambda b)|$$

according to Equation 2.1.

We separate $f((1-\lambda)a + \lambda b) = (1-\lambda)f((1-\lambda)a + \lambda b) + \lambda f((1-\lambda)a + \lambda b)$ and use triangle inequality to get

$$\begin{aligned}
&|((1-\lambda)f(a) + \lambda f(b)) - f((1-\lambda)a + \lambda b)| \\
&\leq |(1-\lambda)f(a) - (1-\lambda)f((1-\lambda)a + \lambda b)| + |\lambda f(b) - \lambda f((1-\lambda)a + \lambda b)| \\
&\leq (1-\lambda)L_f\|a - ((1-\lambda)a + \lambda b)\|_2 + \lambda L_f\|b - ((1-\lambda)a + \lambda b)\|_2 \\
&= 2\lambda(1-\lambda)L_f\|a - b\|_2.
\end{aligned}$$

This completes the proof.

A.3 A Very Looooooooooooooooooooooooooong Title

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

A.4 Some Standard Subsection

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed

iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut.

Appendix B: Something Else I Want to IncludeSome Standard Subsection

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate

aut a voluptate discedere. Nam cum ignoratione rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatibusque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibique, si mihi probabis ea, quae dicta sunt ab iis quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitate aliqua naturam ipsam movet et cum iucunditate quadam.